

LESSON 9
SYMBOLIC ARGUMENTS AND EULER DIAGRAMS

Overview:

In this lesson the different forms of valid and invalid arguments will be discussed and how one can determine whether a given argument is valid or invalid. This lesson will help students on establishing a valid argument out of some given premises.

Learning Outcomes:

- At the end of Lesson 9, the students can:
1. determine whether an argument is valid or not;
 2. illustrate different types of valid and invalid arguments;
 3. establish the validity of an argument; and
 4. use an Euler diagram in showing the validity of an argument.

Materials Needed:

- learning module
- video lesson
- slide presentation

Duration: 3 hours

Learning Content

9.1 ARGUMENT

An **argument** consists of a set of statements called **premises** and another statement called **conclusion**. An argument is **valid** if the conclusion is true whenever all the premises are assumed to be true. An argument is **invalid** if it is not a valid argument.

Let the statements $P_1, P_2, P_3, \dots, P_n$ be the premises of the argument, and q is the conclusion. Arguments can be written in symbolic form:

$$\begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_n \\ \hline \therefore q \end{array}$$

Consider the argument: “If my alarm clock sounds, then I will wake up. My alarm clock sounded. Therefore, I woke up.”. We can write the argument as follows.

<i>First Premise, P₁:</i>	If my alarm clock sounds, then I will wake up.
<i>Second Premise, P₁:</i>	My alarm clock sounded.
<i>Conclusion, q:</i>	Therefore, I woke up.

The above argument can be written in symbolic form. For instance, if we let *c* be the statement “My alarm clock sounds.” and *w* for the statement “I will wake up.”, then the argument can be expressed as

$$\begin{array}{l} c \rightarrow w \\ \underline{c} \\ \therefore w \end{array}$$

Example

- Write the following argument in symbolic form.
The weather is good or I will not come. The weather is good. Therefore, I will come.

Solution

Let *g* represent “*The weather is good.*” and *c* for “*I will come.*”. The symbolic form of the argument is

$$\begin{array}{l} g \vee \sim c \\ \underline{g} \\ \therefore c \end{array}$$

9.2 ARGUMENTS AND TRUTH TABLES

An argument can be determined to be valid using a truth table. Below is the truth table procedure.

Truth Table Procedure to Determine the Validity of an Argument

- Express the argument in symbolic form.
- Construct the truth table showing the truth value of each premise and the truth value of the conclusion for all possible truth value of the simple statements.
- If the conclusion is true in every row of the truth table in which all the premises are true, the argument is valid. If the conclusion is false in every row in which all the premises are true, the argument is invalid.

Example

2. Once again, let us consider the argument “If my alarm clock sounds, then I will wake up. My alarm clock sounded. Therefore, I woke up.”. Determine the validity of the argument using a truth table.

Solution

Step 1

We let,

c : My alarm clock sounds.

w : I will wake up.

The symbolic argument is expressed as follows.

$c \rightarrow w$	First Premise
c	Second Premise
$\therefore w$	Conclusion

Step 2

We are going to construct the truth table.

c	w	First Premise $c \rightarrow w$	Second Premise c	Conclusion w	
T	T	T	T	T	row 1
T	F	F	T	F	row 2
F	T	T	F	T	row 3
F	F	T	F	F	row 4

Step 3

The first row is the only row which contains true on all the premises, and so this is the only row that we examine. Because its conclusion is also true, thus the argument is valid.

3. Consider the following argument from a doctor.

If you take my medication, then you will get better. You are not taking my medication. Therefore, you will not get better.

Solution

Let, m : You take my medication.

t : You will get better.

Hence, its symbolic form is

$m \rightarrow t$	First Premise
$\sim m$	Second Premise
$\therefore \sim t$	Conclusion

The following table is its truth table.

m	t	First Premise $m \rightarrow t$	Second Premise $\sim m$	Conclusion $\sim t$	
T	T	T	F	F	row 1
T	F	F	F	T	row 2
F	T	T	T	F	row 3
F	F	T	T	T	row 4

Because the conclusion in row 3 is false and the premises are both true, the argument is invalid.

4. Determine whether the following argument is valid or invalid.

If you are a fire-eater, then you work in the circus.

If you don't like cotton candy, then you don't work in the circus.

$\therefore$ Therefore, if you are a fire-eater, then you like cotton candy.

Solution

Label each of the simple statements.

f : You are a fire-eater.

w : You work in the circus.

c : You like cotton candy.

The symbolic form of the argument is

$f \rightarrow w$ First Premise

$\sim c \rightarrow \sim w$ Second Premise

$\therefore f \rightarrow c$ Conclusion

Construct the truth table.

f	w	c	First Premise $f \rightarrow w$	Second Premise $\sim c \rightarrow \sim w$	Conclusion $f \rightarrow c$	
T	T	T	T	T	T	row 1
T	T	F	T	F	F	row 2
T	F	T	F	T	T	row 3
T	F	F	F	T	F	row 4
F	T	T	T	T	T	row 5
F	T	F	T	F	T	row 6
F	F	T	T	T	T	row 7
F	F	F	T	T	T	row 8

The only rows containing true on both premises are 1, 5, 7, and 8. Because the conclusion is true in each of these rows, the argument is valid.

9.3 STANDARD FORMS

An argument can be shown to be valid if it has the same symbolic form as an argument that is known to be valid.

The following tables present the standard forms of four valid arguments and standard forms of two invalid arguments.

Table 9.1 Standard Forms of Four Valid Arguments

Direct Reasoning	Contrapositive Reasoning	Transitive Reasoning	Disjunctive Reasoning	
$p \rightarrow q$ $\underline{p}$ $\therefore q$	$p \rightarrow q$ $\underline{\sim q}$ $\therefore \sim p$	$p \rightarrow q$ $\underline{q \rightarrow r}$ $\therefore p \rightarrow r$	$p \vee q$ $\underline{\sim p}$ $\therefore q$	$p \vee q$ $\underline{\sim q}$ $\therefore p$

Table 9.2 Standard Forms of Two Invalid Arguments

Fallacy of the Converse	Fallacy of the Inverse
$p \rightarrow q$ $\underline{q}$ $\therefore p$	$p \rightarrow q$ $\underline{\sim p}$ $\therefore \sim q$

Example

Use a standard form to determine a valid conclusion for each argument.

1. “If the baby is crying (c), then the baby is hungry (h)”
“The baby is not hungry ($\sim h$).”
 $\therefore ?$

Solution

The symbolic form of the premises is:

- $c \rightarrow h$ First Premise
- $\sim h$ Second Premise

This matches the standard form known as *contrapositive reasoning*. Thus a valid conclusion is $\sim c$: “The baby is not crying.”.

2. “If it is the last month of the year (m), there are many items on sale (s).”
“If there are many items on sale (s), I can buy more (b).”
 $\therefore ?$

Solution

The symbolic form of the premises is:

$m \rightarrow s$ First Premise

$s \rightarrow b$ Second Premise

This matches the standard form known as *transitive reasoning*. Thus a valid conclusion is $m \rightarrow b$: "If it is the last month of the year, I can buy more."

Determine whether the argument is valid or not using the standard forms of valid and invalid arguments.

1. The internet connection is fast or I will go to the library.

The internet connection is slow.

$\therefore$ I will go to the library.

Solution

Label the simple statements.

f : The internet connection is fast.

l : I will go to the library.

The symbolic argument is as follows.

$$\begin{array}{c} f \vee l \\ \sim f \\ \hline \therefore l \end{array}$$

This matches one of the standard forms known as *disjunctive reasoning*. Hence the argument is valid.

2. If I ride on a roller coaster, I get dizzy.

I am dizzy.

$\therefore$ I ride on a roller coaster.

Solution

Label the simple statements.

c : I ride on a roller coaster.

d : I get dizzy.

The symbolic argument is as follows.

$$\begin{array}{c} c \rightarrow d \\ d \\ \hline \therefore c \end{array}$$

This matches one of the standard forms known as *fallacy of the converse*. Hence the argument is invalid. Riding on a roller coaster is not the only cause of dizziness. For instance, you may feel dizzy because of too much hunger.

9.4 SEQUENCE OF VALID ARGUMENTS

Constructing a truth table in showing the validity of an argument can be time consuming. For instance, an argument with four simple statements requires $2^4 = 16$ rows and a lot of truth values to be determined. This can be a nightmare to some and we can make an error along the way.

Using a sequence of valid arguments will help us on determining whether an argument is valid or not. Below is the summary of equivalent statements. Some of these are already discussed on the previous lessons.

Table 9.3

Let p , q , and r be statements and we denote tautologies by τ and self-contradictions by ϕ .

Logical Equivalences

Identity Laws	$(p \wedge \tau) \equiv p$	or	$(p \vee \phi) \equiv p$
Domination Laws	$(p \vee \tau) \equiv \tau$	or	$(p \wedge \phi) \equiv \phi$
Idempotent Laws	$(p \vee p) \equiv p$	or	$(p \wedge p) \equiv p$
Inverse Laws	$[p \vee (\sim p)] \equiv \tau$	or	$[p \wedge (\sim p)] \equiv \phi$
Double Negation	$\sim (\sim p) \equiv p$		
Associative Laws	$p \vee (q \vee r) \equiv (p \vee q) \vee r$	or	$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$
Commutative Laws	$p \vee q \equiv q \vee p$	or	$p \wedge q \equiv q \wedge p$
Distributive Laws	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	or	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
De Morgan's Laws	$\sim (p \vee q) \equiv \sim p \wedge \sim q$	or	$\sim (p \wedge q) \equiv \sim p \vee \sim q$
Absorption Laws	$p \vee (p \wedge q) \equiv p$	or	$p \wedge (p \vee q) \equiv p$
Switcheroo	$p \rightarrow q \equiv \sim p \vee q$		

Example

1. Consider the symbolic argument below and determine whether the argument is valid or invalid using a sequence of valid arguments.
- $a \rightarrow b$

$b \rightarrow c$

a

(Premise 1)

(Premise 2)

(Premise 3)

$\therefore c$

(Conclusion)

Solution

We assume that all simple statements above the line are true. From these three statements (premises), we want to arrive at the conclusion c using a sequence of valid arguments.

Statement		Reason
1	$a \rightarrow b$	Premise 1
2	$b \rightarrow c$	Premise 2
3	$a \rightarrow c$	(1) and (2), Transitive Reasoning
4	a	Premise 3
5	c	(3) and (4), Direct Reasoning

Since the last row of the Statement column is c by *direct reasoning* and is exactly the same with the given conclusion, the argument is valid.

2. Determine if the following argument is valid:

$m \rightarrow n$	(Premise 1)
$\sim n \vee r$	(Premise 2)
$r \rightarrow \sim s$	(Premise 3)
s	(Premise 4)
$\therefore \sim r$	(Conclusion)

Solution

Statement		Reason
1	$\sim n \vee r$	Premise 2
2	$n \rightarrow r$	(1), Switcheroo
3	$m \rightarrow n$	Premise 1
4	$r \rightarrow \sim s$	Premise 3
5	$m \rightarrow \sim s$	(3), (2), and (4), Transitive Reasoning
6	s	Premise 4
7	$\sim m$	(5) and (6), Contrapositive Reasoning

Using the sequence of valid arguments above, it shows that the conclusion is $\sim m$ and is not logically equivalent to $\sim r$. Hence, the argument is invalid.

3. Determine if the following argument is valid:
- If you want to pass this subject, then you have to work hard for it. Your final grade must be greater than 50% or you did not work hard for it. Your final grade is less than 50%. Therefore, you did not pass the subject.

Solution

We label the arguments as follows.

- p : You want to pass this subject.
- w : You have to work hard for it.
- g : Your final grade must be greater than 50%.

In symbolic form, the argument is

$$\begin{array}{ll} p \rightarrow w & \text{(Premise 1)} \\ g \vee \sim w & \text{(Premise 2)} \\ \sim g & \text{(Premise 3)} \\ \hline \therefore \sim p & \text{(Conclusion)} \end{array}$$

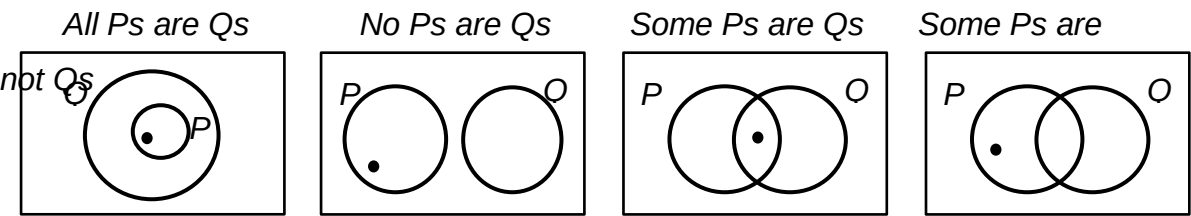
Solution

Statement		Reason
1	$g \vee \sim w$	Premise 2
2	$w \rightarrow g$	(1), Switcheroo
3	$p \rightarrow w$	Premise 1
4	$p \rightarrow g$	(2) and (3), Transitive Reasoning
5	$\sim g$	Premise 3
6	$\sim p$	(4) and (5), Contrapositive Reasoning

Using the sequence of valid arguments above, it shows that the conclusion is $\sim p$. Hence, the argument is valid.

9.5 EULER DIAGRAMS

An Euler diagram is used in determining if an argument involving quantifiers is valid or invalid



Example

4. Use an Euler diagram to determine whether the given argument is valid or not.

All college courses are fun.
This course is a college course.
 $\therefore$ This course is fun.

Solution

The first premise indicates that the set of college courses is a subset of the set of fun courses. We illustrate this subset relationship with an Euler diagram, as show in Figure 9.1. The second premise tells us that “this course” is an element of the set of college students. If we use c to represent “this course”, then c must be placed inside the set of college courses, as shown in Figure 9.2.

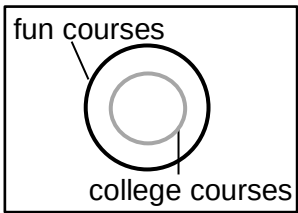


Figure 9.1

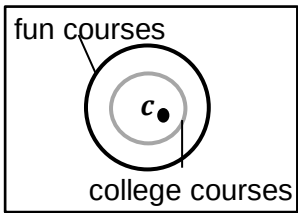


Figure 9.2

5. Use an Euler diagram to determine whether the following argument is valid or invalid.

No psychologist can juggle.
All clowns can juggle.
 $\therefore$ No psychologist is a clown.

Solution

Figure 9.3 shows that the two sets are disjoint. However, since the set of clowns is a subset of the set of jugglers as shown in Figure 9.4, no psychologists p are elements of the set of clowns. This the argument is valid.

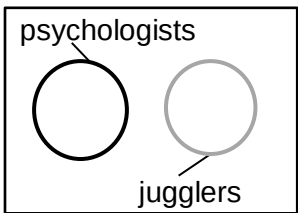


Figure 9.3

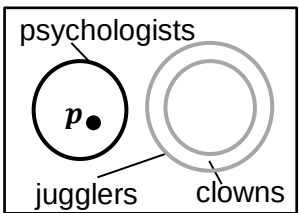


Figure 9.4

Learning Activity

Think-Pair-Share:

Each pair will be assigned to answer two problems (Problem A and Problem B).
Problem A. Let students work by pairs. Each pair will pick an argument by lottery. Student A will determine the validity of the argument by means of a truth table, while student B will determine the validity of the same argument by using a sequence of valid arguments. Students A and B will compare their outputs by examining one’s work.

Problem B. Working with the same pair, each will determine the validity of the given argument using an Euler diagram.

Pair 1:

- A. If I plant a tree, then I will get dirt under my nails. I didn't get dirt under my nails. Therefore, I didn't plant a tree.
- B. All horses have hooves. Some horses eat oats. Therefore, some oat-eaters have hooves.

Pair 2:

- A. If I don't change my oil regularly, my engine will die. My engine died. Thus, I didn't change my oil regularly.
- B. Some mammals are bats. All bats can fly. Thus, some mammals can fly.

Pair 3:

- A. You will meet a tall, handsome stranger or you will stay home and pick fleas off of your cat. You didn't meet a tall, handsome stranger. Therefore, you stayed home and picked fleas off of your cat.
- B. No dogs can talk. Some searchers are dogs. Thus, some searchers can't talk.

Pair 4:

- A. If I don't tie my shoes, then I trip. I didn't tie my shoes. Hence, I tripped.
- B. Some squirrels fly. Some squirrels gather acorns. Therefore, some acorn-gatherers fly.

Pair 5:

- A. If I lose my keys, then I can't start my car. If I lose my keys, then I can't get in my house. Therefore, if I can't start my car, then I can't get in my house.
- B. Some scavengers eat road-kill. All crows eat road-kill. Therefore, all crows are scavengers.

Pair 6:

- A. If you aren't polite, you won't be treated with respect. You aren't treated with respect. Therefore, you aren't polite.
- B. All successful politicians are persuasive. Some lawyers are persuasive. Thus, some lawyers are successful politicians.

Pair 7:

- A. If you are kind to a puppy, then he will be your friend. You weren't kind to that puppy. Hence, he isn't your friend.
- B. Some bikers drink Lipovitan. Some bikers drink Gatorade. Therefore, some Lipovitan drinkers are Gatorade drinkers.

Pair 8:

- A. If you drink Lipovitan, then you won't fall off of your skateboard. You fell off of your skateboard. Therefore, you didn't drink Lipovitan.
- B. All tattooers are body-artists. Some tattooers drive Big Bike. Therefore, some body-artists drive Big Bike.

Pair 9:

- A. If I don't pay my income taxes, then I file for an extension or I am a felon. I'm not a felon and I didn't file for an extension. Therefore, I paid my income taxes.
- B. Some multiple-choice questions are tricky. No easy questions are tricky. Therefore, some multiple-choice questions are not easy.

Pair 10:

- A. If the power goes off, then the air conditioner will not work. The air conditioner is working. Therefore, the power is not off.
- B. Some snails live under rocks. All snails are slimy. Therefore, some things that live under rocks are slimy.

Learning Evaluation:

- A. Determine the validity of the following arguments using truth tables.
 - 1. I wash the dishes or I don't eat. I eat. Thus, I wash the dishes.
 - 2. If I am literate, then I can read and write. I can read but I can't write. Thus, I am not literate.
 - 3. If it rains or snows, then my roof leaks. My roof is leaking. Thus, it is raining and snowing.
 - 4. If I lose my keys, then I can't start my car. If I lose my keys, then I can't get in my house. Therefore, if I can't start my car, then I can't get in my house.
 - 5. If it rains or snows, then my roof leaks. My roof is leaking. Thus, it is raining and snowing.
- B. Determine the validity of each argument using a sequence of valid arguments.
 - 1. If you want to be a used-car salesman, then you have to be a flashy dresser. You don't want to be a used-car salesman. Thus, you don't have to be a flashy dresser.
 - 2. If an animal is cute, then it isn't a squid. This animal isn't a squid. Therefore, this animal is cute.
 - 3. If you play golf during a thunderstorm, you'll get hit by lightning. You didn't get hit by lightning. Therefore, you didn't play golf during a thunderstorm.

4. I will run for office or I will shut my mouth. I ran for office. Thus, I didn't shut my mouth.
 5. If I am literate, then I can read and write. I can read but I can't write. Thus, I am not literate.
- C. Determine the validity of each argument using a sequence of Euler diagrams.
1. All astronauts are bold. Some pilots are not bold. Therefore, some astronauts are not pilots.
 2. Some ants are aggressive. All ants are insects. Therefore, some insects are aggressive.
 3. All plumbers use monkey wrenches. Some mechanics don't use monkey wrenches. Hence, some mechanics aren't plumbers.
 4. All spies are secretive. Some agents aren't spies. Therefore, some agents aren't secretive.
 5. Some lawyers are judges. Some judges are politicians. Therefore, some lawyers are politicians.

References:

Aufman, R. et. al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc. Sampaloc, Manila, Philippines.

Johnstone, P. (1987). *Mathematics, Logic, Categories, and Sets*. Cambridge University Press.